Project Description

02268 – Process-oriented and Event-driven Software Systems

Last Update: September 23, 2026

General instructions

You will implement a process-oriented application system including connection with event sources and a process aware information system (PAIS). You will identify and model processes AS-IS, and improve them making sure they have appropriate notions of quality. You will refine processes AS-IS, finding ways to improve their initial behaviour, making them ready to be executable in a PAIS. You will operate processes and generate event-logs, that you can audit using notions of compliance in DCR graphs.

General requirements:

- The project must be conducted in teams of 6 persons.
- The project consists of identifying a scenario relevant in terms of processes automation and event generation/consumption, implementing it and generating logs for analysis.
- The process implementation should use Process Aware Information Systems (PAIS) (e.g., Camunda **and** DCR graphs).
- The project should make use of a system for Complex Event Processing (CEP) (e.g., Siddhi, Apache Flink).
- The project must include interaction between PAIS and CEP **in both directions** (i.e., from the CEP platform to the PAIS and from the PAIS to the CEP platform).
- Each group may choose their own scenario based on their experience, interests, working experience or, general preference but there is the possibility to use one of the themes/scenarios suggested in this document.
- Each group must identify the in their application domain, select a group of processes for deeper modelling and implementation, select event sources and implement their application.
- Your project must be unique, that means, you can share a theme, but the processes modelled and implemented will need to be unique to each group.

To be successful, the project must involve the following *minimum requirements*:

- 6 business processes (at least one model must be a BPMN model, and one model should be a DCR graph[1] where most of elements discussed during the lecture are properly used). Please mark in the hand in the student responsible for each process model. Each student should develop at least one process model.
- At least one process modeled should correspond to the Knowledge-Intensive Process category.

[1] You can use DCR graphs or Petri nets, or other formalisms, for the remaining processes, if the group considers appropriate.

- Implementation of the processes in at least 1 PAIS using Camunda, and 1 PAIS using DCR graphs (*at least* means that 1 is enough)
- 2 external event sources
- Some processing of the events with proper CEP agents (e.g., Siddhi or Apache Flink)
- **Bi-directional communication** between the PAIS and the CEP.
- One set of compliance rules for each of the implemented process models (modelled as DCR graphs)
- Analysis of compliance checking of the executions via event-log generation and conformance checking (using processdoctor.dtu.dk)

The deliverable should include a project report where the group describes what they have done in a critical way (i.e., motivate their decision) as well as provide a link to a URL with an exhaustive screencast of all implementation details; the code of the application itself, and the slides of the group presentation. Further instruction below.

Suggested Themes

Suggested theme 1: Disaster Management Processes

We live in a world that is in constant change. The effects of climate change are affecting countries, that need to react promptly to floods[2] or earthquakes[3]. Political instability may require cities to evacuate due to bombings and other unexpected situations. Multiple organizations are involved in a disaster management process. From firemen that need to extinguish fires and police officers that need to secure an area, to insurance funds that need to assess damage claims and pay claims.

Your task in this theme will be 1) to identify one organization involved in disaster management processes, 2) identify its core processes associated to proper incident response, 3) model event sources (e.g. sensors identifying earth movement, water flow, etc). 4) propose a set of automated and semi-automated processes that can help supporting endangered population in a prompt manner.

Suggested theme 2: Improved public transportation system

The public transportation system (for example busses and trains) is a very structured and well-orchestrated system where a lot of events need to happen at precise steps, to have it working smoothly. In this context, different notions of processes are possible.

[2] https://en.wikipedia.org/wiki/2026_Nepal%E2%80%93Tibet_floods
[3] https://en.wikipedia.org/wiki/2026_Colombia_earthquake



For example, one could imagine a bus route as a sequential process, where each activity is a bus stop that should be visited. The progress of such a process is typically automatically handled (i.e., when a bus reaches a stop, a certain event is triggered and the process is moved forward) and many other require synchronization (for example, when busses approaches “connected bus stops” and need to wait for other busses to arrive before continuing).

Aside from normal operations, other type of processes could also be investigated. For example, after a certain amount of km/operation hours, maintenance should be scheduled. Other factors affecting the operation may involve external conditions (e.g. weather) or internal (e.g. vandalism). Alternatively, if busses are getting delayed too much (i.e., bottlenecks), new busses might be dispatched. In other cases, the interconnection of multiple means of transportation may trigger changes in the schedules and routes (for instance, if the metro service is disrupted, its operation may be supported via metrobuses).

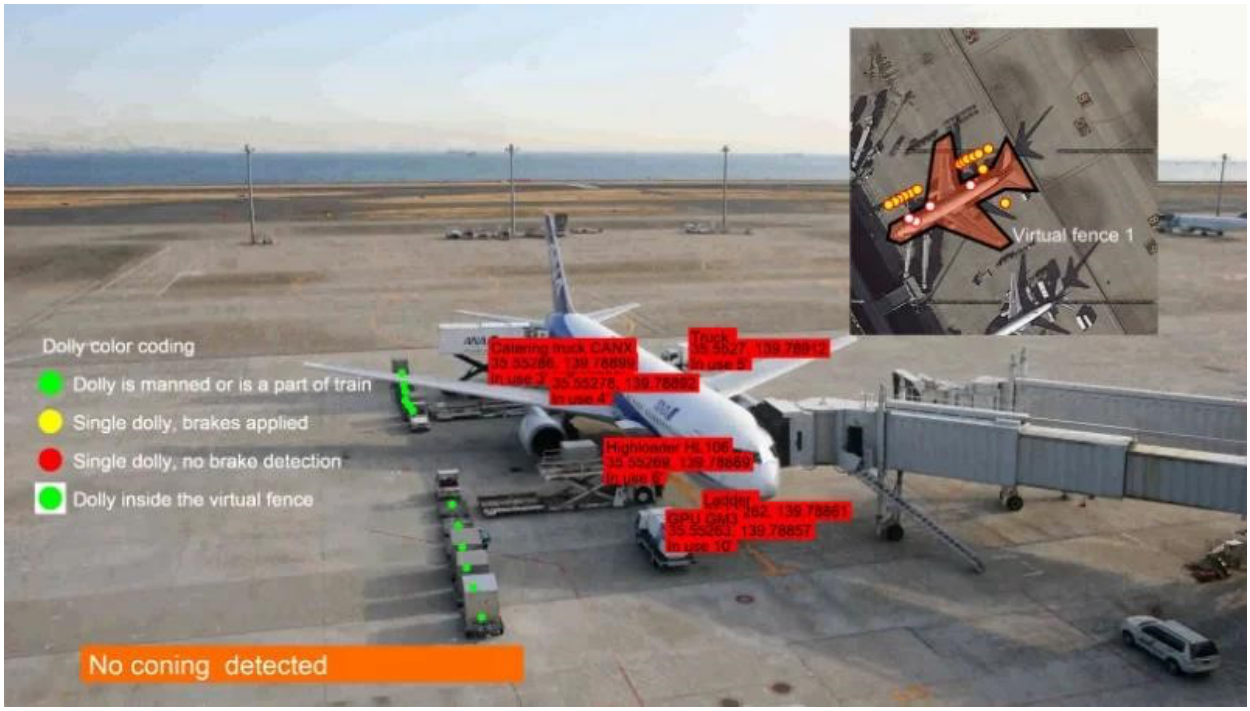
In Denmark, DOT[4] serves as a connection between different transportation companies and such interconnection allows the gathering of data, which is then consumed, for example by Rejseplanen in their live visualization of all busses, trains (tog and S-tog), and metro[5]. The company also offers access to a set of REST APIs that can be consumed to know live information on trains[6].

Suggested theme 3: Process tracking from video stream

Many real-world processes are “observed” via cameras. For example, consider the turnaround process at an airport[7]. Improving the digitalization of these processes could result in tremendous

[4] See https://dinoffentligetransport.dk/en/about-dot/.
[5] See https://www.rejseplanen.dk/bin/help.exe/mn?L=vs_dot.vs_livemap&tpl=fullscreenmap.
[6] See https://help.rejseplanen.dk/hc/en-us/articles/214174465-Rejseplanen-s-API and also a BSc project on the subject https://findit.dtu.dk/en/catalog/6083fcacd9001d01701fe6b7.
[7] See https://findit.dtu.dk/en/catalog/5a1dfd1c5010df2563769897.

cost saving for airports and airlines and, as a result, for passenger's tickets. Initiatives to achieve part of such digitalization are in place[8].



In a recent Master project[9], two students worked on consuming data from cameras to understand how the turnaround process progresses. Integrating data from multiple sources (such as video + ADB-S[10]) could help improving the quality of the digitalization and hence the connection with the expected turnaround process.

Other domains might also be explored. For example, at DTU there is an initiative called DTU Smart Library[17] which aims at exposing data from sensors already deployed in the library.

Suggested theme 4: Predictive maintenance

Predictive maintenance is defined as

> *A condition-driven preventive maintenance program. Instead of relying on industrial or in-plant average-life statistics (i.e., mean-time-to-failure) to schedule maintenance activities, predictive maintenance uses direct monitoring of the mechanical condition, system efficiency, and other indicators to determine the actual mean-time-to-failure or loss of efficiency for each machine-train and system in the plant.* [11]

[8] See https://assaia.com/turnaround-control.
[9] See https://findit.dtu.dk/en/catalog/6216242583fe41235e4fb8af.
[10] See https://en.wikipedia.org/wiki/Automatic_Dependent_Surveillance%E2%80%93Broadcast. [17]
See https://www.bibliotek.dtu.dk/en/visit-the-library/dtu-smart-library.
[11] From Section 1.1.3 of "An Introduction to Predictive Maintenance", https://findit.dtu.dk/en/catalog/2305333571.



In case you are working or have knowledge of a specific facility or equipment, it might be possible to develop a predictive maintenance strategy by exploiting data being generated by sensors and analyzed using CEP. Such knowledge would then trigger specific maintenance processes with the goal of preserving the production capacity. ‘

Suggested theme 5: Talent Management Processes



This project will be as close to reality as possible. Universities runs multiple processes including teaching, research, innovation, and management. Part of the pledges of the university is to show transparency in its hiring and career progression processes, in this way illustrating that decisions being taken are inclusive and merit based.
This project will aim at 1. Discover the most important HR processes included in the talent management of personnel (e.g. via inspiration on DTU Compute processes), and 2. Illustrate alternatives on how processes could be improved, and 3. Develop a simulator for a case management system who could support the HR administration in the management of hirings and progression plans.
For this project you will interface with real stakeholders, and you will get support for the scheduling of interviews to gather/understand the status quo of existing processes. Your CEP

events will simulate possible situations in the workplace (e.g. new job applications, changes of role, increase number of absences, reviews, etc.).

Key dates

Project announcement:	22 September 2026
Group and topic decision:	12 October 2026- on DTU Learn, same as assignment groups
Intermediate presentation:	3 November 2026
Submission of all material:	15 December 2026 23:59
Final presentation:	TBA (in agreement with censor) - exact timing and scheduling will be provided once groups are known

What to submit

Before the deadline, **each group must submit a `.zip` file** containing three elements. Please, **be precise with the naming of each element**:

1. A report in a file called **`report.pdf`** (the report must be in PDF) which should be structured according to the document `demo-report.pdf` (attached to this project specification):
   - Must be maximum 10 pages long (spare the space: do not use one page just for the title, another for the table of contents, etc.)
   - Must provide a complete description of the context, the processes being used, why the group chose to implement them. In addition, a complete description of the events and corresponding processing must be provided in detail
   - Must contain a link to a screencast (max 3 minutes) where **all** implemented functionalities must be shown and commented.
2. The complete source code in a file called **`source.zip`**, including:
   - Process definitions (as `.bpmn` or DCR `xml` file as well as `.pdf`)
   - CEP agents source codes
   - Implementations of services (e.g., in Java), if any
   - PAIS customizations, if any.
3. The final presentation in a file called **`presentation.pdf`** (the presentation must be exported in PDF).

Checklist of topics covered

To make sure that your project achieves the learning objectives, you can use this checklist and verify that all corresponding elements are discussed in your report and presentations:

	✓
I have applied methods and techniques for analyzing business processes to identify improvement opportunities	
I have identified and created business process models using a business process modeling notation like BPMN, DCR graphs or Petri nets	
I have improved early designs of business processes based on different redesign heuristics	
I have implemented a business process directly from a business process model using a PAIS	
I have discussed different approaches to integrated business process management and sensor events	
I have integrated event processing with a business process management system	
I have established communications from event sources to a business process management system	
I have established communications from a business process management system to IoT devices (or simulations via event streams)	
I have worked in team to solve a problem	

Checklist of submission material

To make sure that your project respects the rules of the submission, you can use the following checklist:

	✓
The submission comprises just one zip file	
The zip file contains one document called report.pdf which is ≤ 10 pages long	
The zip file contains a file called source.zip with all process files and all other sources	
The zip file contains one document called presentation.pdf	

The intermediate presentation

There will be a first mandatory group presentation where each group has a certain number of minutes (around 10/15 minutes, exact time will follow once I know how the number of groups) to present the status of their project (+ 3 minutes for questions/answers). The goals of this presentation are:

- Practice with presentation and dealing with time constraints as well as questions (that is why it is important to carefully prepare this presentation)
- Receive feedback from instructors as well as from other students (that is why it is important to carefully prepare this presentation)
- Get inspiration from other groups' work (that is why it is important to attend and be involved in the discussions).

This presentation **is not** part of the grading (it is up to you get the most out of this opportunity).

The final presentation

Each group will need to present their project in a 10-minute-long presentation. During the presentation, it is required that:

- Each member of the group presents something.
- The presentation has not to be a "slides" version of the report. The focus must be on the elements that are better explained in person. Still, it must cover all elements of the project, including
  - Domain description
  - Process landscape and processes implementation
  - Events sources considered and how these have been processed
  - How the interaction between events and processes took place
  - Challenges, take-home messages, and conclusions
- Each member must be able to answer questions about any part of the project
- The presentation might include a brief live demonstration of the system if you think this can contribute to the better understanding of the project (e.g., in case a phone is generating events, maybe you want to demonstrate this).
- After the presentation, questions will be asked to each group member, starting from the project and possibly covering theoretical aspects as well. The timing of the question phase will be announced once precise information about the group is available.

Assessment

The final assessment is based on the group work; and the individual assignment (released separately).

Appendix: examples of APIs with events

If you decide not to work with the suggested themes, here are some examples of free API describing possible events, which can be used for the project:

- OpenSky REST API: https://openskynetwork.github.io/opensky-api/
- AIS Hub: http://www.aishub.net/api
- Rejseplanen real-time data: https://help.rejseplanen.dk/hc/da/categories/201728005 (and also https://findit.dtu.dk/en/catalog/6083fcacd9001d01701fe6b7)
- Twitter real-time Tweets: https://developer.twitter.com/en/docs/twitter-api/tweets/filteredstream/introduction
- Current weather conditions: https://openweathermap.org/current
- Facebook web-hooks: https://developers.facebook.com/docs/pages/realtime/
- GitHub events: https://developer.github.com/v3/activity/events/
- Sports events: https://apifootball.com/documentation/ and https://allsportsapi.com/ • Environmental Portal: https://arealinformation.miljoeportal.dk/html5/index.html?viewer=distribution

- Open Data Denmark: https://www.opendata.dk/
- Live energy production and consumption in Bornholm, https://bornholm.powerlab.dk/ and https://bornholm.powerlab.dk/visualizer/latestdata
- Researchers' life via UI tracking: https://github.com/project-pivot/labelled-awt-data
- Log generator: https://link.springer.com/chapter/10.1007/978-3-031-70396-6_13